

Yen-Hsun LIN

Curriculum Vitae

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RESEARCH SUMMARY

I am an astroparticle physicist studying dark matter (DM) and new particle interactions through supernovae and stellar environments. I pioneered supernova-neutrino-boosted DM, showing how its delayed arrival-time signature can probe the DM mass and extending the analysis to past Galactic supernovae and the cumulative signal across cosmic history. I also proposed using the circumstellar medium of supernovae to search for feebly interacting particles through the heating and optical emission induced by their decays. Building on my work on DM capture, thermal transport, and compact-star heating, I am developing a kinetic Boltzmann-transport framework to follow captured DM beyond thermal equilibrium. This framework aims to quantify how capture, scattering, evaporation, and annihilation shape the DM distribution and its effects on stellar objects, with applications to DM-powered stars and the formation of compact dark objects inside neutron stars.

Research Interests

Astrophysical probes of DM and feebly interacting particles; supernova neutrinos and electromagnetic transients; DM capture, transport, and heating in stars; numerical methods, Monte Carlo simulation, and Bayesian inference.

Programming and tools

Agentic AI, Python, C++, Mathematica, and MATLAB.

EDUCATION

National Chiao Tung University

PhD, Institute of Physics

Thesis: Indirect detection of dark matter through neutrinos

Advisor: Prof. Guey-Lin Lin

Hsinchu, Taiwan
Aug. 2011 – Jun. 2016

National Chiao Tung University

Master's degree, Institute of Physics (direct admission to the PhD program)

Advisor: Prof. Guey-Lin Lin

Hsinchu, Taiwan
Aug. 2010 – Jul. 2011

National Chiao Tung University

Bachelor's degree, Department of Electrophysics

Hsinchu, Taiwan
Aug. 2006 – Jul. 2010

EXPERIENCE

NSTC Research Scholar

Institute of Physics, Academia Sinica
PI of an NSTC Research Grant

Taipei, Taiwan
Aug. 2026 – Present

Postdoctoral Scholar

Institute of Physics, Academia Sinica
Host: Dr. Meng-Ru Wu

Taipei, Taiwan
Aug. 2023 – Jul. 2026

Visiting Scholar

School of Physics, Melbourne University
Host: Prof. Nicole F. Bell

Melbourne, Australia
Oct. 2023 – Nov. 2023

Postdoctoral Scholar

Physics Division, National Center for Theoretical Sciences

Taipei, Taiwan
Dec. 2021 – Jul. 2023

Distinguished Postdoctoral Scholar

Institute of Physics, Academia Sinica
Host: Dr. Meng-Ru Wu

Taipei, Taiwan
Aug. 2019 – Dec. 2021

Postdoctoral Researcher

Department of Physics, National Cheng Kung University
Host: Prof. Chuan-Hung Chen

Tainan, Taiwan
Oct. 2017 – Jul. 2019

HONORS & AWARDS

1. **Shui-Chin Lee Fellow** Taiwan, 2026
Selected by the Shui-Chin Lee Foundation.
2. **Postdoctoral Academic Research Award** Taiwan, 2024
Awarded by the Division of Particles and Fields, Taiwan Physical Society.
3. **NCTS Postdoc Paper Award** Taiwan, 2024
Awarded by the Physics Division, National Center for Theoretical Sciences (NCTS).
4. **Best Research Paper Award for Junior Research Investigator** Taiwan, 2024
Awarded by the Institute of Physics, Academia Sinica.
5. **Selected Participant of the 13th HOPE Meeting with Nobel Laureates** Japan, 2022
Representative of Taiwan.
6. **Distinguished Postdoctoral Scholar** Taiwan, 2019
Independent position with grant, awarded by Academia Sinica.
7. **Postgraduate Student Thesis Award** Taiwan, 2017
Selected as the best PhD thesis of the year in physical sciences, awarded by the Taiwan Physical Society.
8. **Selected Honorary Member of the Phi Tau Phi Scholastic Society** Taiwan, 2016
Awarded to the graduating student with the highest academic standing.

RESEARCH GRANTS

1. **PI of an NSTC Research Grant** 2026 – 2029
Investigating dark matter distribution function in stellar objects with the Boltzmann transport equation. Grant No. 115-2112-M-001-037-MY3
2. **AS Postdoctoral Fellowship with Research Grant** 2019 – 2021
Compact stellar object as a new probe on particle dark matter.

COLLABORATION MEMBERSHIP

1. Collaboration with members of the DUNE and COHERENT Collaborations USA
 - ◊ Collaborating with Dr. Gianluca Petrillo and Dr. Yun-Tse Tsai 2020 – 2024
 - ◊ Analyzing the impact of ν_e -Ar cross-section uncertainties
 - ◊ Improving sensitivity to pinched spectral parameters using machine learning
2. Jiangmen Underground Neutrino Observatory (JUNO) Jiangmen, China
 - ◊ Co-author of the JUNO Yellow Book (R&D technical notes) 2015 – 2016
 - ◊ Sensitivity projection for solar-captured DM at JUNO

ADVISEES

PhD students

1. **Vo Quang Nhat** Hsinchu, Taiwan
Institute of Physics, National Yang Ming Chiao Tung University Aug. 2022 – Jul. 2023
Co-supervised with Prof. Guey-Lin Lin
2. **Lam Thi To Uyen** Hsinchu, Taiwan
Institute of Physics, National Yang Ming Chiao Tung University Aug. 2022 – Jul. 2023
Co-supervised with Prof. Guey-Lin Lin

Undergraduates

1. **Hsuan Lee** (NSTC Research Assistant) Hsinchu, Taiwan
Department of Physics, National Tsing Hua University Jul. 2026 – Present
Supported by NSTC Grant No. 115-2112-M-001-037-MY3.
2. **Gabrielle Roque** (NSTC-IIPP Program) Manila, Philippines
B.S. in Applied Physics, University of Santo Tomas Jul. 2025 – Aug. 2025
Co-supervised with Dr. Meng-Ru Wu. The student was sponsored by the International Internship Pilot Program (IIPP) of the National Science and Technology Council (NSTC), Taiwan.
3. **Tsung-Han Tsai** (ASIoP Summer Student Program) Hsinchu, Taiwan
Department of Physics, National Tsing Hua University Jul. 2022 – Oct. 2023
Co-supervised with Dr. Meng-Ru Wu; the work was published in *Phys. Rev. D* **108**, 083013 (2023). The student is currently a PhD student at the University of Wisconsin–Madison (USA).
4. **Yong Sheng Yap** (ASIoP Summer Student Program) Hsinchu, Taiwan
Department of Physics, National Tsing Hua University Jul. 2021 – Aug. 2021
Co-supervised with Dr. Meng-Ru Wu. The student is currently a PhD student at the University of Cambridge (UK).
5. **Wen-Hua Wu** (ASIoP Summer Student Program) Taipei, Taiwan
Department of Physics, National Taiwan University Jul. 2020 – Mar. 2023
Co-supervised with Dr. Meng-Ru Wu; the work was published in *Phys. Rev. Lett.* **130**, 111002 (2023). The student is currently a PhD student at Rice University (USA).
6. **Adeela Malik** (SLAC Summer Student Program) San Antonio, USA
Department of Physics, University of Texas at San Antonio Jul. 2020 – Aug. 2020
Co-supervised with Prof. Hirohisa Tanaka (SLAC) and Dr. Yun-Tse Tsai (SLAC).

SCIENTIFIC ACTIVITIES & SERVICES

Workshop organization

- Organizer of the Mini-workshop on Novel Experimental and Astrophysical Probes for Dark Matter, Taipei, Taiwan, 2021.

Journal referee

1. Physics Letters B
2. Annals of Physics

Publication List

REFEREED ARTICLES

Dagger (†) and asterisk (*) indicate *first author* and *corresponding author*, respectively. Otherwise, the author list is arranged alphabetically.

1. Y. Cheng, C.-F. Kong, **Y.-H. Lin**^{*}, M.-R. Wu, and S. Yun, *Circumstellar medium of supernovae as new probes for feebly-interacting particles*, *Phys. Rev. Lett.* **137**, 111005 (2026), [[arXiv:2603.09615](#)]
2. **Y.-H. Lin**^{†,*} and M.-R. Wu, *Mapping the evolution of supernova-neutrino-boosted dark matter within the Milky Way*, *Phys. Rev. D* **112**, 083001 (2025) [[arXiv:2506.15151](#)]
3. **Y.-H. Lin**^{†,*} and M.-R. Wu, *Supernova-neutrino-boosted dark matter from all galaxies*, *Phys. Rev. Lett.* **133**, 111004 (2024) [[arXiv:2404.08528](#)]
4. **Y.-H. Lin**^{†,*}, T.-H. Tsai, G.-L. Lin, H. T.-K. Wong and M.-R. Wu, *Signatures of afterglows from light dark matter boosted by supernova neutrinos in current and future large underground detectors*, *Phys. Rev. D* **108**, 083013 (2023) [[arXiv:2307.03522](#)]
5. **Y.-H. Lin**^{†,*}, W.-H. Wu, M.-R. Wu and H. T.-K. Wong, *Searching for afterglow: Light dark matter boosted by supernova neutrinos*, *Phys. Rev. Lett.* **130**, 111002 (2023) [[arXiv:2206.06864](#)]
6. A. Bauswein, G. Guo, J.-H. Lien, **Y.-H. Lin** and M.-R. Wu, *Compact dark objects in neutron star mergers*, *Phys. Rev. D* **107**, 083002 (2023) [[arXiv:2012.11908](#)]
7. G.-L. Lin and **Y.-H. Lin**^{*}, *Exploring dark sector parameters in light of neutron star temperatures*, *Phys. Rev. D* **104**, 063021 (2021) [[arXiv:2102.11151](#)]
8. G.-L. Lin and **Y.-H. Lin**^{*}, *Analysis on the black hole formations inside old neutron stars by isospin-violating dark matter with self-interaction*, *JCAP* **08**, 022 (2020) [[arXiv:2004.05312](#)]
9. C.-S. Chen and **Y.-H. Lin**^{*}, *Reheating neutron stars with the annihilation of self-interacting dark matter*, *JHEP* **08**, 069 (2018) [[arXiv:1804.03409](#)]
10. C.-S. Chen and **Y.-H. Lin**^{*}, *On the evolution process of two-component dark matter in the Sun*, *JHEP* **04**, 074 (2018) [[arXiv:1802.06956](#)]
11. C.-H. Chen and **Y.-H. Lin**, *Study of $B_c^\pm \rightarrow (D^0 K^\pm, D^0 \pi^\pm)$ decays*, [[arXiv:1710.05531](#)]
12. C.-S. Chen, G.-L. Lin, **Y.-H. Lin** and F. Xu, *The 17 MeV anomaly in beryllium decays and $U(1)$ portal to dark matter*, *Int. J. Mod. Phys. A* **32**, 1750178 (2017) [[arXiv:1609.07198](#)]
13. C.-S. Chen, G.-L. Lin and **Y.-H. Lin**, *Thermal transport of the solar captured dark matter and its impact on the indirect dark matter search*, *Phys. Dark Univ.* **14**, 35 (2016) [[arXiv:1508.05263](#)]
14. C.-S. Chen, G.-L. Lin and **Y.-H. Lin**, *Complementary test of the dark matter self-interaction by direct and indirect detections*, *JCAP* **01**, 013 (2016) [[arXiv:1505.03781](#)]
15. G.-L. Lin, **Y.-H. Lin** and F.-F. Lee, *Probing the coupling of heavy dark matter to nucleons by detecting neutrino signature from the Earth core*, *Phys. Rev. D* **91**, 033002 (2015) [[arXiv:1409.3094](#)]
16. C.-S. Chen, F.-F. Lee, G.-L. Lin and **Y.-H. Lin**, *Probing dark matter self-interaction in the Sun with IceCube-PINGU*, *JCAP* **10**, 049 (2014) [[arXiv:1408.5471](#)]

COLLABORATION PAPERS

1. A. Abusleme *et al.* [JUNO Collaboration], *Measurement of reactor neutrino oscillation with the first JUNO data*, *Nature* **654**, 343 (2026) [[arXiv:2511.14593](#)]
2. A. Abusleme *et al.* [JUNO Collaboration], *Initial performance results of the JUNO detector*, *Chin. Phys. C* **50**, 043001 (2025) [[arXiv:2511.14590](#)]
3. A. Abusleme *et al.* [JUNO & Daya Bay Collaborations], *Optimization of the JUNO liquid scintillator composition using a Daya Bay antineutrino detector*, *Nucl. Instrum. Meth. A* **988**, 164823 (2021) [[arXiv:2007.00314](#)]
4. Z. Djurcic *et al.* [JUNO Collaboration], *JUNO conceptual design report*, [arXiv:1508.07166](#)
5. F. An *et al.* [JUNO Collaboration], *Neutrino Physics with JUNO*, *J. Phys. G* **43**, 030401 (2016) [[arXiv:1507.05613](#)]

PREPRINTS

1. Y. Cheng, C.-F. Kong, Y.-H. Lin, M.-R. Wu, and S. Yun, *Probing feebly interacting particles with 511 keV line from circumstellar medium of supernovae*, (2026), [[arXiv:2609.18247](#)]
2. Y.-H. Lin, M.-R. Wu, and S. Yun, *Dark matter evolution in stellar objects: A Boltzmann transport approach*, *To be submitted*

CONFERENCE PROCEEDINGS

1. G.-L. Lin and **Y.-H. Lin**, *Exploring dark sector parameters in light of neutron star temperatures*, *PoS ICHEP2022* 106 (2022)
2. Lam T. T. Uyen, G.-L. Lin and **Y.-H. Lin**, *Constraints on lepton-flavor-violating scalar portal using the Belle II result in the search for $e^+e^- \rightarrow e^\pm + \text{invisible}$ with $L = 276 \text{ pb}^{-1}$* , *PoS ICHEP2022* 1229 (2022)
3. **Y.-H. Lin** and G.-L. Lin, *Analysis on the black hole formations inside old neutron stars by isospin-violating dark matter with self-interaction*, *PoS ICHEP2020* 598 (2020)
4. **Y.-H. Lin** and G.-L. Lin, *Probing self-interacting dark matter through neutron stars*, *PoS EPS-HEP2019* 075 (2020)
5. C.-S. Chen, G.-L. Lin **Y.-H. Lin** and F. Xu, *The 17 MeV anomaly in beryllium decays and $U(1)$ portal to dark matter*, *PoS EPS-HEP2017* 627 (2017)
6. C.-S. Chen, **Y.-H. Lin** and G.-L. Lin, *Complementary test of the dark matter self-interaction in dark $U(1)$ model by direct and indirect detection*, *JPS Conf.Proc.* **14**, 020103 (2017)
7. **Y.-H. Lin**, C.-S. Chen and G.-L. Lin, *Thermal transport of the solar captured dark matter and its implication*, *JPS Conf.Proc.* **14**, 020110 (2017)
8. C.-S. Chen, G.-L. Lin and **Y.-H. Lin**, *Probing dark matter self-interaction in the Sun with IceCube-PINGU*, *PoS FPC2015*, 065 (2015)
9. **Y.-H. Lin**, C.-S. Chen and G.-L. Lin, *Thermal transport of the solar captured dark matter and its implication*, *PoS EPS-HEP2015*, 385 (2015)
10. C.-S. Chen, F.-F. Lee, G.-L. Lin and **Y.-H. Lin**, *The dark matter self-interaction and its impact on the critical mass for dark matter evaporations inside the Sun*, *Nucl.Part.Phys.Proc.* **273-252**, 347-375 (2016)

Presentations

INVITED CONFERENCE/WORKSHOP TALKS

1. *Supernova-neutrino-boosted dark matter: From the next galactic SN to all SNe in the Universe*
The 4th Workshop on Symmetry and Structure of the Universe, Jeonju, Korea (2025/8)
2. *Exploring light dark matter boosted by supernova neutrinos in the past and present Universe*
Particle Physics Phenomenology Workshop (PPP 15), Taipei, Taiwan (2024/10)
3. *Detecting afterglow signatures from light dark matter boosted by supernova neutrinos*
Interstellar and Intergalactic Insights: Exploring the Energetic Universe with Multi-messengers, Taipei, Taiwan (2023/12)
4. *Detection of $SN\nu$ BDM in current and future large underground detectors*
The 3rd International Joint Workshop on the Standard Model and Beyond and the 11th KIAS Workshop on Particle Physics and Cosmology, Jeju, Korea (2023/11)
5. *Searching for afterglow: Light dark matter boosted by supernova neutrinos*
Interplay of Nuclear, Neutrino and BSM Physics at Low-Energies (INT 23-85w), Seattle, USA (2023/4)
6. *Light DM constraints from neutron stars and supernova neutrinos*
Theory Meets Experiment: Particle Physics and Cosmology, Quy Nhon, Vietnam (2023/1)
7. *Light DM constraints from neutron stars and supernova neutrinos*
NCTS Annual Theory Meeting, Taipei, Taiwan (2022/12)
8. *Light dark matter boosted by supernova neutrinos*
Dark Matter in Compact Objects, Stars, and in Low Energy Experiments (INT 22-2b), Seattle, USA (2022/8)
9. *Searching the afterglow from supernova neutrino boosted dark matter*
Particle Physics Phenomenology Workshop (PPP 14), Taipei, Taiwan (2022/6)
10. *Exploring dark matter with compact stars*
Mini-workshop on Novel Experimental and Astrophysical Probes for Dark Matter, Taipei, Taiwan (2021/3)
11. *Probing the isospin violation of self-interacting dark matter through old neutron stars*
NCTS Annual Theory Meeting, Taipei, Taiwan (2019/12)
12. *Probing self-interacting dark matter through neutron stars*
Particle Physics Phenomenology Workshop (PPP 13), Taipei, Taiwan (2019/6)

CONTRIBUTED CONFERENCE/WORKSHOP TALKS

1. *Supernovae as versatile probes of physics beyond the Standard Model*
54th International Symposium on Multiparticle Dynamics (ISMD 2026), Chungli, Taiwan (2026/8)
2. *Circumstellar medium of SNe as new probes for feebly-interacting particles*
Particle Physics Phenomenology Workshop (PPP 16), Hsinchu, Taiwan (2026/6)

3. *Shimmering darkness: Mapping the distribution of supernova-neutrino-boosted dark matter across the sky*
Neutrino and Dark Matter 2025 (NDM 2025), Paris, France (2025/5)
4. *Exploring light dark matter boosted by supernova neutrinos in the present and past Universe*
International Joint Workshop on the Standard Model and Beyond & the 3rd Gordon Godfrey Workshop on Astroparticle Physics, Sydney, Australia (2024/12)
5. *Searching light dark matter boosted by supernova neutrinos in Super-K, Hyper-K and DUNE*
International Conference on Topics in Astroparticle and Underground Physics (TAUP 2023), Vienna, Austria (2023/8)
6. *Detection of afterglows from supernova-neutrino boosted dark matter in large underground detectors*
International Summer Institute on Phenomenology of Elementary Particle Physics and Cosmology, Nantou, Taiwan (2023/8)
7. *Analysis on the black hole formations inside old neutron stars by isospin-violating dark matter with self-interaction*
International Conference on High Energy Physics (ICHEP 2020), Prague, Czech Republic (2020/8)
8. *The art of inference: Practicing Bayesian reasoning in computer vision problems*
PyCon TW, Taipei, Taiwan (2019/9)
9. *Probing self-interacting dark matter through neutron stars*
European Physical Society Conference on High Energy Physics (EPS-HEP 2019), Ghent, Belgium (2019/7)
10. *On the evolution process of two-component dark matter in the Sun*
International Conference on Neutrino Physics and Astrophysics (Neutrino 2018), Heidelberg, Germany (2018/6)
11. *Thermal transport of the solar captured dark matter and its implication*
International Symposium on Nuclei in the Cosmos (NIC-XIV), Niigata, Japan (2016/6)
12. *Thermal transport of the solar captured dark matter and its implication*
European Physical Society Conference on High Energy Physics (EPS-HEP 2015), Vienna, Austria (2015/7)
13. *The dark matter self-interaction and its impact on the critical mass for dark matter evaporations inside the Sun*
International Conference on High Energy Physics (ICHEP 2014), València, Spain (2014/7)
14. *Probing the coupling of heavy dark matter to nucleons by detecting neutrino signature from the Earth core*
International Symposium on Particles, Strings and Cosmology (PASCOS 2014), Taipei, Taiwan (2013/12)
15. *Probing the coupling of heavy dark matter to nucleons by detecting neutrino signature from the Earth core*
International Symposium on Cosmology and Particle Astrophysics (CosPA 2013), Honolulu, Hawaii (2013/6)

SEMINARS & COLLOQUIA

1. *Circumstellar medium of supernovae as new probes for feebly-interacting particles*
Seminar for the Institute of Physics, National Yang Ming Chiao Tung University, Taiwan (2026/5)
2. *Behind the veil of darkness: A journey into the Uncharted Universe*
Colloquium for the Department of Physics, Tunghai University, Taichung, Taiwan (2024/5)

3. *Exploring light dark matter boosted by supernova neutrinos in the present and past Universe*
Seminar for the Department of Physics, National Tsing Hua University, Hsinchu, Taiwan (2024/5)
4. *Detecting afterglow signatures from light dark matter boosted by supernova neutrinos*
Seminar for the Department of Physics, Chinese University of Hong Kong, Hong Kong (2024/1)
5. *Signatures of afterglows from light dark matter boosted by supernova neutrinos in large underground detectors*
Seminar for the Theoretical Particle Physics (TPP) Group, School of Physics, Melbourne University, Melbourne, Australia (2023/10)
6. *Search for afterglow: Light dark matter boosted by supernova neutrino*
Seminar for the Department of Physics, National Tsing Hua University, Hsinchu, Taiwan (2023/5)
7. *Search for afterglow: Light dark matter boosted by supernova neutrino*
Seminar for the Department of Physics, Chung Yuan Christian University, Taoyuan, Taiwan (2023/3)
8. *Search for afterglow: Light dark matter boosted by supernova neutrino*
Webinar for the Supernova Early Warning System (SNEWS), USA (2023/2)
9. *Search for afterglow: Light dark matter boosted by supernova neutrino*
Seminar for the Department of Physics, National Taiwan Normal University, Taipei, Taiwan (2022/11)
10. *Searching for afterglow: Light dark matter boosted by supernova neutrinos*
NCTS Particle Physics Journal Club, National Taiwan University, Taipei, Taiwan (2022/9)
11. *Searching light to heavy dark matter by supernova neutrinos and neutron star temperature*
Webinar for the Sydney Consortium for Particle Physics and Cosmology (Sydney-CPPC), Sydney, Australia (2021/10)
12. *Probing dark matter with neutron star*
Seminar for the Department of Physics, Chung Yuan Christian University, Taoyuan, Taiwan (2020/11)
13. *Probing dark matter with neutron star*
Seminar for the Center of Astrophysics and Gravity, National Taiwan Normal University, Taipei, Taiwan (2020/7)
14. *Neutron star sensitivity on isospin-violating dark matter with self-interaction*
Seminar for the Department of Physics, National Central University, Taoyuan, Taiwan (2020/6)
15. *Uncharted Universe*
Colloquium for the Department of Physics, Tamkang University, New Taipei, Taiwan (2018/12)
16. *Reheating neutron stars with the annihilation of self-interacting dark matter.*
Seminar for the Institute of Physics, National Chiao Tung University, Hsinchu, Taiwan (2018/5)
17. *Reheating neutron stars with the annihilation of self-interacting dark matter.*
Seminar for the Institute of Physics, Academia Sinica, Taipei, Taiwan (2018/4)
18. *Indirect detection of dark matter through neutrinos*
Seminar for the Department of Physics, Chung Yuan Christian University, Taoyuan, Taiwan (2017/12)